Masruk Habib

Software Engineer · Data & BI · Automation

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PROFILE

Computer engineer who builds software, data and automation that people rely on. At GoScurry.ai I design and ship features end to end on PostgreSQL, FastAPI and React: schema changes and migrations, SQL and vector search, integrations with outside systems, and the automated tests behind them. Before that I spent eight months as a data analyst at Jatri in Dhaka, turning sales, finance and supply chain data into Metabase dashboards and automated reports and keeping the numbers consistent across systems. I like to understand why a system was built the way it was, and I explain technical work plainly in English and Bangla, through five published papers, 37 teaching sessions and a technical blog. Now looking to grow into technology leadership in financial services.

EXPERIENCE

Junior Software Engineer · GoScurry.ai — **Apr 2026 to present**

Chiang Mai, Thailand · Remote · AI meeting intelligence platform, React 19 and TypeScript on FastAPI and PostgreSQL

- **Scope:** second-largest contributor on the team, with 366 merged pull requests and roughly 140,000 lines of production code across frontend, backend, database schema and CI.
- **Databases and SQL:** wrote 49 Alembic migrations evolving the PostgreSQL schema, and built the RAG personalisation layer on pgvector with HNSW indexing, retrieval logging for tuning, and a batch pipeline that keeps bulk embedding affordable.
- **System design:** own customer onboarding end to end through two full rewrites, covering 22 backend modules and an eight-screen React flow that validates and atomically commits an executable workflow graph for each account.
- **Integration and data mapping:** designed the Pipedrive CRM integration, with AI-ranked field shortlisting that narrows 49 fields to 10, field descriptions generated from real data, and mapping profiles between the two systems.
- **Testing and controls:** wrote 223 automated test files (pytest, Vitest) that run in CI, and added a per-field human review gate so nothing is written back to a customer's CRM unchecked.
- **Billing and security:** shipped the billing and growth systems behind spend caps, usage metering, low-balance alerts and referral commissions, and authored the prompt injection defence applied to every untrusted transcript, email body and fetched page.

Data Analyst · Jatri — **Sep 2025 to Apr 2026**

Dhaka, Bangladesh

- Analysed sales, marketing, finance and supply chain data in Metabase to answer the questions the business actually had.
- Built the dashboards and visualisations and automated the recurring reporting, so numbers arrived without anyone chasing them.
- Checked and kept data consistent across systems, and worked with whichever team needed numbers that week.
- Supported demand forecasting and sales planning, flagging risks and opportunities as they appeared.
- Tracked sales growth, inventory and market trends, and how customers responded to promotions.

Backend Developer · CRM Squirrel — **Aug 2025 to Sep 2025**

Lampang, Thailand (remote)

- Wrote Python backend services behind core CRM workflows.
- Handled API integration and helped shape how the system would scale.

Student Research Mentor · Dept. of Computer Engineering, Marwadi University — **Nov 2023 to Apr 2025**

Rajkot, Gujarat, India

- Took three batches of students from first draft through to a published paper.
- Ran 37 sessions at the Research Activity Club, explaining machine learning and how to write results up properly.
- Co-authored two papers with students from other countries.

Research Intern, SRIT · Dept. of Computer Engineering, Marwadi University — **May 2024 to Jun 2024**

Rajkot, Gujarat, India

- Wrote and published one paper over the two months, under a faculty mentor.

EDUCATION

B.Tech in Computer Engineering — **Nov 2021 to Jun 2025**

Marwadi University, Rajkot, Gujarat, India · CGPA 7.64 / 10.00, First Class with Distinction

Higher Secondary (HSC) and Secondary School (SSC), Science — **2018 and 2020**

BIAM Model School and College, Bogura, Bangladesh · GPA 5.00 / 5.00 in both

SKILLS

Databases & SQL	SQL, PostgreSQL, schema design and migrations (Alembic), SQLAlchemy 2, pgvector, Redis, Oracle SQL
Data & BI	Data analysis, data quality checks, Metabase, Tableau, dashboard prototyping, KPI tracking, forecasting, data storytelling, Google Sheets, Google Analytics
Programming	Python, TypeScript and JavaScript, SQL · data structures · FastAPI, REST APIs, OAuth2, Django, Flask · React 19, Vite, Tailwind, Streamlit
Testing & Delivery	pytest, Vitest, Postman, GitHub Actions CI, Docker Compose, Git, Linux, AWS
AI & Automation	Python scripting, automated reporting, LLM integration (Anthropic, OpenAI, xAI, Google), RAG, prompt injection defence, machine learning, anomaly detection, time series (ARIMA, LSTM)
Languages	Bangla (native), English (full professional), Hindi (elementary)

PROJECTS

Tableau dashboards public.tableau.com/app/profile/masruk.habib
Public dashboards covering marketing campaigns, Superstore sales history, and HR analytics and operations.

Portfolio and technical blog masrukhabib.github.io
A Jekyll site where I write technical work up in plain language, from running production RAG on PostgreSQL with pgvector to a beginner's guide to becoming a business analyst.

Phishing website detection github.com/MasrukHabib/Phishing-website-detection-using-machine-learning
A classifier that spots malicious sites from their URL and page features, so someone gets warned before they click.

Multilingual retrieval-augmented generation system github.com/MasrukHabib/Multilingual-Retrieval-Augmented-Generation-RAG-System
Answers a question asked in one language using documents written in another. Embeddings and vector search keep every answer tied back to a real passage instead of invention.

MediGuard AI: drug and side effect predictor with a RAG chat assistant github.com/MediGuardAI
I led this one. A fine-tuned LLM checks a patient's medications for interactions and likely side effects, with a RAG chat assistant on top for follow-up questions. It runs as a Streamlit app.

UR3 CobotOps: predictive maintenance for a collaborative robot
github.com/MasrukHabib/UR3-CobotOps-Enhancing-Cobot-Efficiency-with-ML
ARIMA, LSTM, K-Means and a few classifiers over the UCI cobot dataset, used to catch anomalies while the robot is running and predict failures before they happen.

ASD Predictor: automated Django application github.com/MasrukHabib/ASD_Predictor_ML_Django_App
A Django app for early autism screening, built on SVM, logistic regression, extra trees and decision trees over UCI data. Tuning brought precision and recall to 1.0. Published at WCONF 2024.

CERTIFICATIONS

AWS Academy Graduate, Cloud Foundations (Amazon Web Services, Oct 2023)
Database Programming with SQL (Oracle Academy, Nov 2022)
The Complete Tableau Bootcamp for Aspiring Data Scientists (Infosys Springboard, Mar 2023)
Project Management Foundations (LinkedIn, Jul 2023)
Python Data Structures (Coursera, Aug 2022)
Postman API Fundamentals Student Expert (Postman, Apr 2023)
Digital Security Essentials
CCNAv7: Introduction to Networks (Cisco Networking Academy, Feb 2023)
NDG Linux Essentials (Cisco Networking Academy, Apr 2023)
Explore Machine Learning using Python (Infosys Springboard, Sep 2024)
Programming for Everybody (Coursera, Jul 2022)
Machine Learning with Python: Foundations

AWARDS

Winner, Azadi Ka Amrit Mahotsav Hackathon 2022 · Government of Gujarat **Mar 2023**
Our team won this state-level hackathon for a project the judges thought could work in practice, not just on a slide.

PUBLICATIONS

Machine Learning Classifiers and Hyperparameter Tuning to Enhance Prediction of ASD: An Automated Django Application
2024 2nd World Conference on Communication & Computing (WCONF), IEEE · DOI: 10.1109/WCONF61366.2024.10692172

Predicting Psychological Well-being Using Machine Learning with Hybrid Hyperparameter Tuning Techniques
2024 International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit), IEEE · DOI: 10.1109/GlobalAISummit62156.2024.10947775

A Comparative Study on Hybrid Machine Learning Voting Classifier Models for Alzheimer's Disease Prediction
2023 International Conference on AI for Innovations in Healthcare (ICAIIHI), IEEE · DOI: 10.1109/ICAIIHI57871.2023.10489772

Augmenting Human Capabilities: AI's Impact on Collaborative Workflows

A Roadmap to a Sustainable Future: Empowering Businesses and Consumers Through AI-Driven Green Marketing